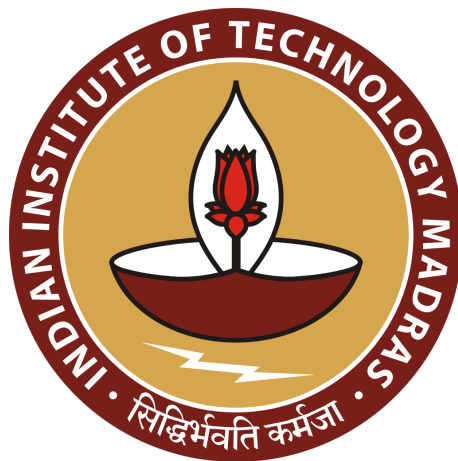


Assignment -1

Visual Statistics and Hypothesis Testing



BT3041 - Analysis and Interpretation of Biological Data

Even Semester 2026

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Question 1

Shown below is the histograms used to visualize the blood glucose distribution for female and male mice across different weeks.

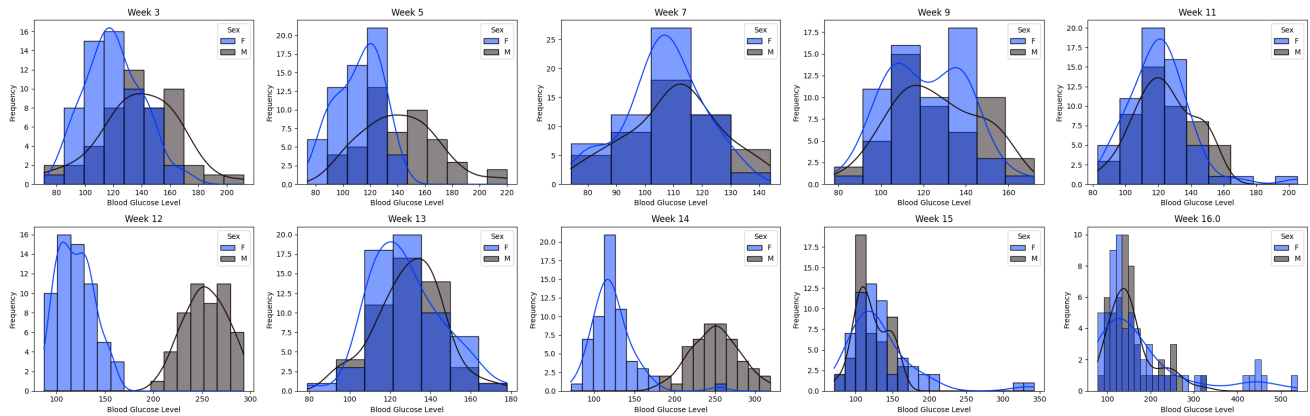


Figure 1: Histograms of blood glucose level at different weeks (Blue - Female; Red - Male)

Interpretation/Analysis:

- Week 3 (Female) - The distribution is symmetric (zero skewness), with low variance.
Week 3 (Male) - The distribution is symmetric (nearly zero skewness), with high variance.
- Week 5 (Female) - The distribution is asymmetric (left skewed), with low variance.
Week 5 (Male) - The distribution is asymmetric (secondary peak), with high variance.
- Week 7 (Female) - The distribution is symmetric (zero skewness), with moderate variance.
Week 7 (Male) - The distribution is symmetric (zero skewness), with moderate variance.
- Week 9 (Female) - The distribution is asymmetric (multiple peaks), with high variance.
Week 9 (Male) - The distribution is asymmetric (presence of a local peak), with high variance.
- Week 11 (Female) - The distribution is symmetric (zero skewness), with low variance.
Week 11 (Male) - The distribution is asymmetric (right skewed), with moderate variance.
- Week 12 - Both the distributions share almost no overlap with each other, which is to be noted.
Week 12 (Female) - The distribution is asymmetric (right skewed), with moderate variance.
Week 12 (Male) - The distribution is symmetric (zero skewness), with moderate variance.
- Week 13 (Female) - The distribution is asymmetric (right skewed), with moderate variance.
Week 13 (Male) - The distribution is asymmetric (zero skewness), with moderate variance.
- Week 14 - Both the distributions similar to week 12 share almost no overlap with each other.
Week 14 (Female) - The distribution is symmetric (zero skewness), with low variance.
Week 14 (Male) - The distribution is symmetric (zero skewness), with moderate variance.
- Week 15 (Female) - The distribution is asymmetric (right skewed), with moderate variance.
Week 15 (Male) - The distribution is asymmetric (local peak), with moderate variance.
- Week 16 (Female) - The distribution is asymmetric (outliers on the right tail), with moderate variance.
Week 16 (Male) - The distribution is asymmetric (presence of local peak), with moderate variance.

Conclusions:

While in general the distributions appear to be symmetric and possess moderate variance (as judged from the histograms), in certain cases as in week 15 and 16, there exist outliers that skew the distribution massively (This can be observed much more clearly in question 2 when looking at the box plots).

Overall, these distributions largely pass the visual (“eye-test”) assessment and can be considered (crudely to be) normally distributed. In certain cases, such as weeks 5 and 9 where multiple peaks appear, this pattern may simply be an artifact of the limited sample size. A larger sample would be required before drawing any definitive conclusions.

Additionally, to further validate these observations, histograms can be examined in conjunction with boxplots, which provide complementary information about the spread of the data and the presence of potential outliers, thereby offering a more informative assessment of the underlying distribution.

Question 2

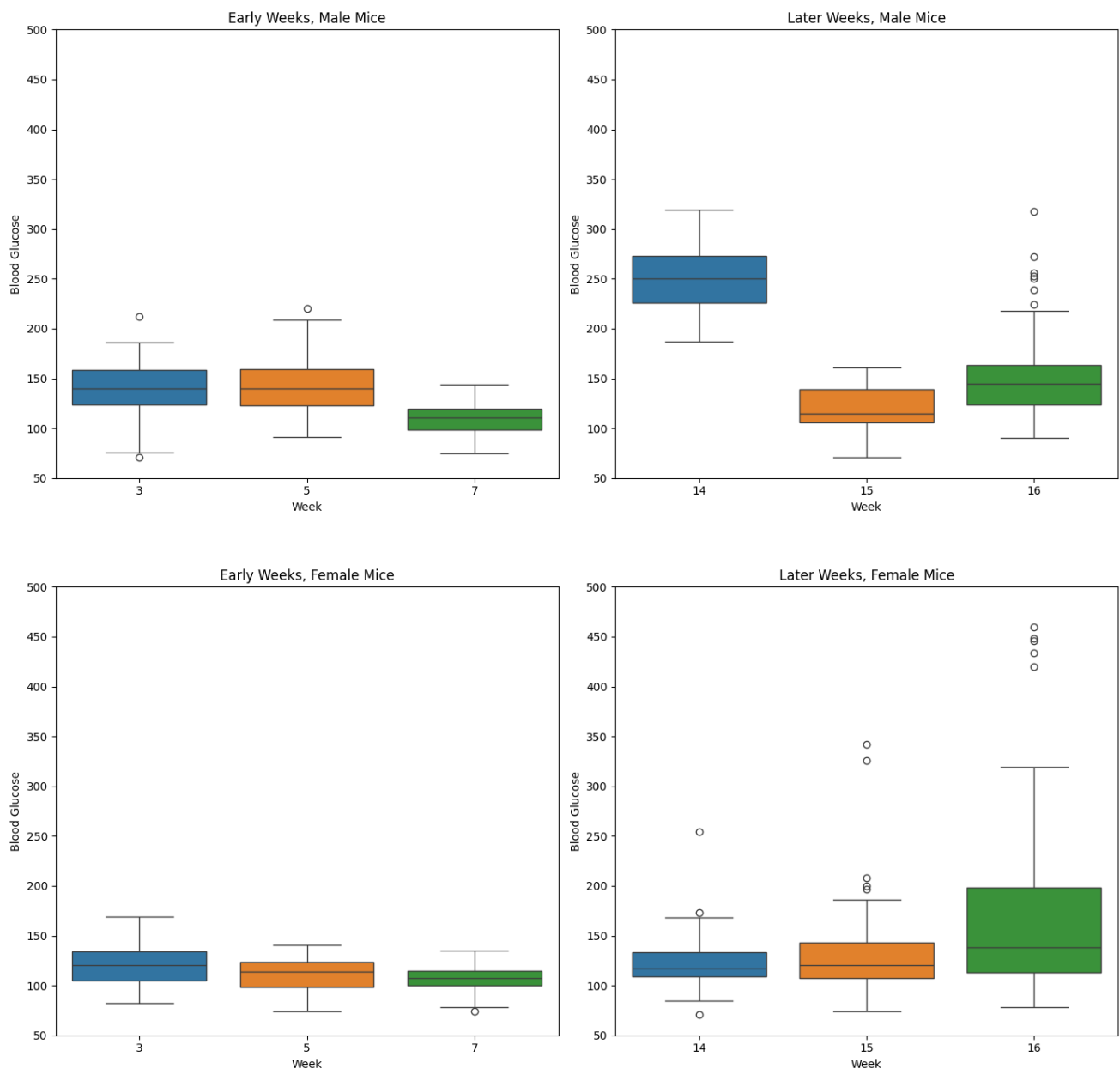


Figure 2: Boxplots illustrating blood glucose distributions of male and female mice across early and later ages

Male Mice:

i) Median (denoted by Q_2):

- In the early phase, median blood glucose of week 3 $\approx$ week 5 $>$ week 7.
- In the late phase, median blood glucose of week 14 $>$ week 16 $>$ week 15.
- Overall week 14 $>$ week 16 $>$ week 3 $\approx$ week 5 $>$ week 15 $>$ week 7.

ii) Interquartile range (denoted by $Q_3 - Q_1$)

In the early phase:

- The interquartile ranges for week 3 and 5 are nearly identical, indicating that the two weeks exhibit very similar distributions.
- week 7 shows a noticeably narrower interquartile range, suggesting a more concentrated distribution around the median (the sharper peak and lower variance is also clearly reflected in the histogram presented in Question 1).

In the late phase:

- The interquartile range of week 14 is moderately sized, indicating a distribution with moderate variance.
- The interquartile range of week 15 is also moderately large; however, more importantly it is asymmetric with respect to the median, suggesting that the distribution is right-skewed.
- Week 16 exhibits an interquartile range similar to that of week 14, indicating a comparable level of variability in the data.

iii) Outliers (denoted by points outside the boxplot)

- In the early phase, there are very few outliers in the week 3 and week 5 data, and the observed outlier points lie relatively close to the main distribution.
- In the late phase, as noted in Question 1, week 16 exhibits several outliers extending toward the far right tail. This results in a skewed distribution, which is captured by the boxplot shown above.

Female Mice:

i) Median (denoted by Q_2):

- In the early phase, median blood glucose of week 3 $>$ week 5 $>$ week 7.
- In the late phase, median blood glucose of week 16 $>$ week 15 $>$ week 14.
- Overall week 16 $>$ week 15 $\approx$ week 3 $>$ week 14 $>$ week 5 $>$ week 7.

ii) Interquartile range (denoted by $Q_3 - Q_1$)

In the early phase:

- The interquartile range of week 3 is moderately sized, indicating that the distribution exhibits moderate variance.
- The interquartile range of week 5 is also moderately sized; however, the median is asymmetric, suggesting that the distribution is left-skewed.
- week 7 shows a noticeably narrower interquartile range, suggesting a more concentrated distribution around the median.

In the late phase:

- The interquartile range of week 14 is narrow, suggesting a sharp peak. Additionally the median is asymmetric, suggesting that the distribution is right-skewed.
- The interquartile range of week 15 is moderately sized, however, the median is asymmetric, suggesting that the distribution is right-skewed.
- The interquartile range of week 16 is large, indicating a broad peak (or multiple peaks). Additionally the median is asymmetric, suggesting that the distribution is right-skewed (This lends more credence to the fact that the distribution probably has multiple local peaks).

iii) Outliers (denoted by points outside the boxplot)

- In the early phase, week 7 has one outlier located close to the main distribution.
- In the late phase, as noted in Question 1, week 14, 15 and 16 exhibits outliers extending toward the far right tail, within week 16 having several outliers far from the main distribution. This results in a skewed distribution, which is captured by the boxplot shown above.

Question 3

i) Across mice within each individual week

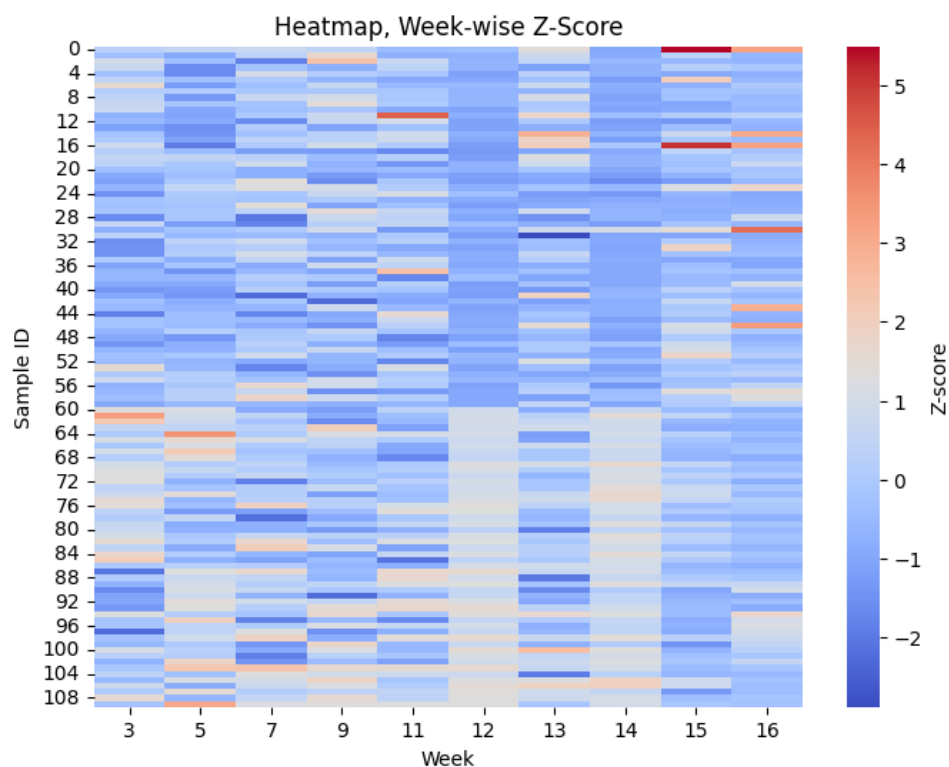


Figure 3.1: Computing a Z-score for each week (Column-wise Z scores)

- This heat map can be used to compare blood glucose levels of different individuals in a given week.
- In general both male and female mice at the same age seem have similar blood glucose levels.
- Weeks 12 and 14 stand out, with male mice showing Z-scores between 1 and 2 and female mice between -1 and -2, consistent with the near-complete separation of their distributions observed in the histograms from Question 1.
- In weeks 15 and 16 in the female samples, bright red bands indicating strong outliers are seen.

ii) Across weeks within each individual mouse.

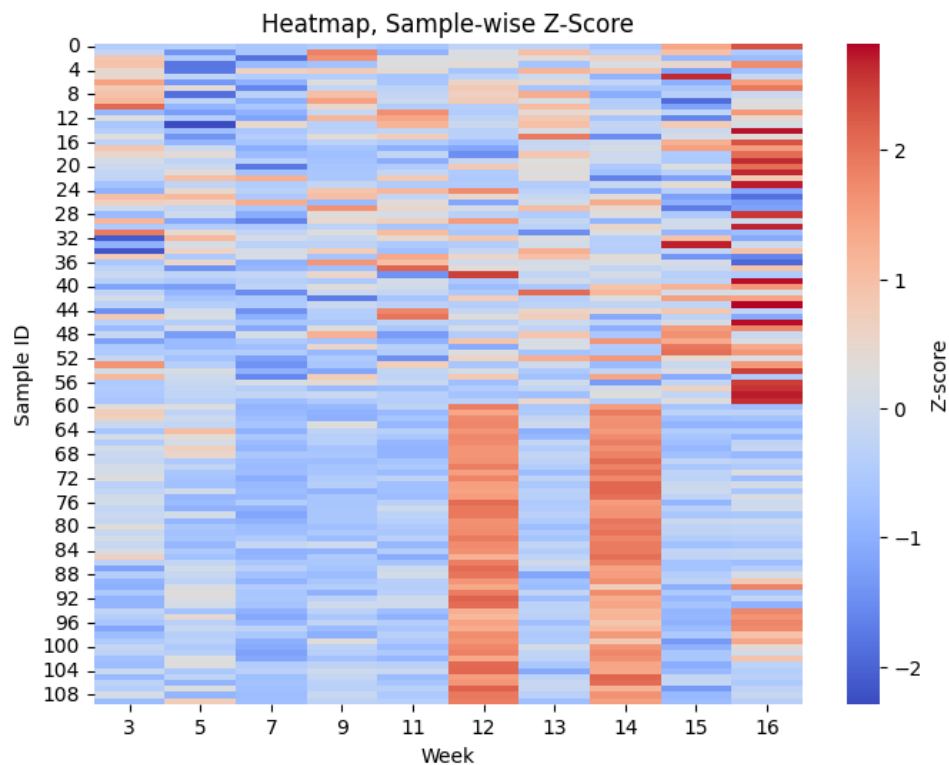


Figure 3.2: Computing Z-score across a sample (Row-wise Z scores)

- This heat map can be used to analyse blood glucose levels of an individuals over time.
- In general, male mice appear to exhibit more consistent blood glucose levels over their lifespan, whereas female mice display greater variability, with no clear overall trend.
- However, weeks 12 and 14 appear as exceptions again as the blood glucose level of male mice spike to much higher values compared to other time points (indicated by the bright red cells).
- Additionally, week 16 appears as an outlier with most female and same male mice having high blood glucose levels.

Question 4

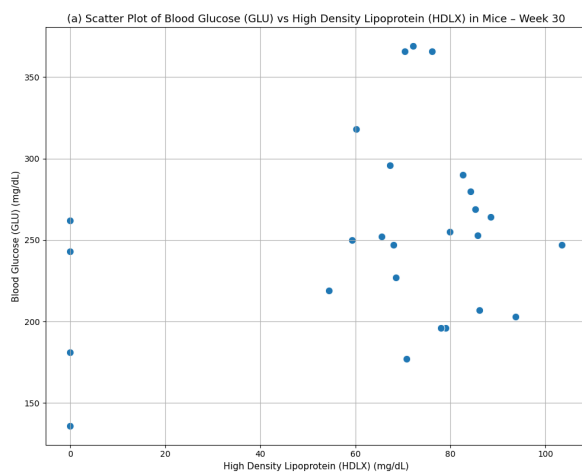


Figure 4.1: GLU vs HDLX (cov = 490.60)

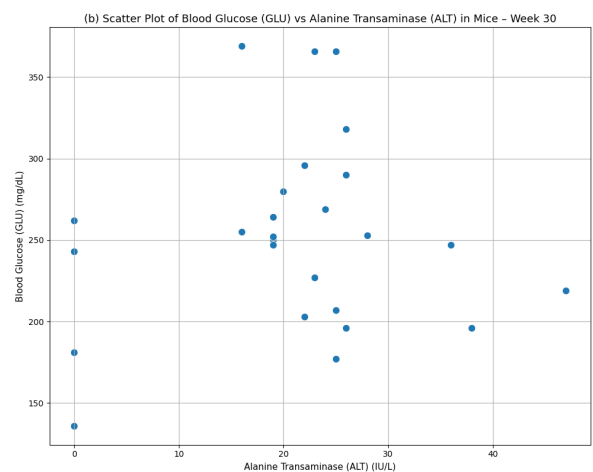


Figure 4.2: GLU vs ALT (cov = 73.25)

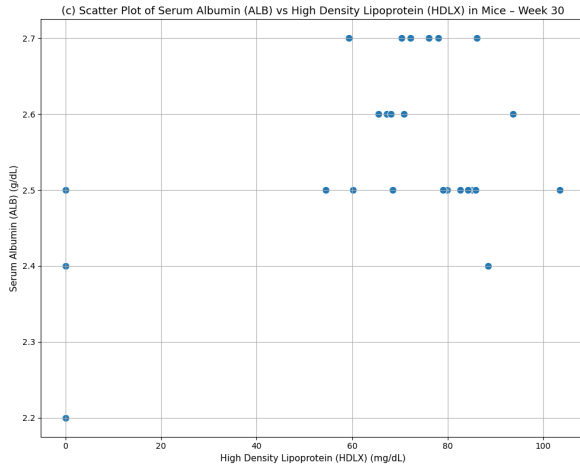


Figure 4.3: ALB vs HDLX (cov = 1.80)

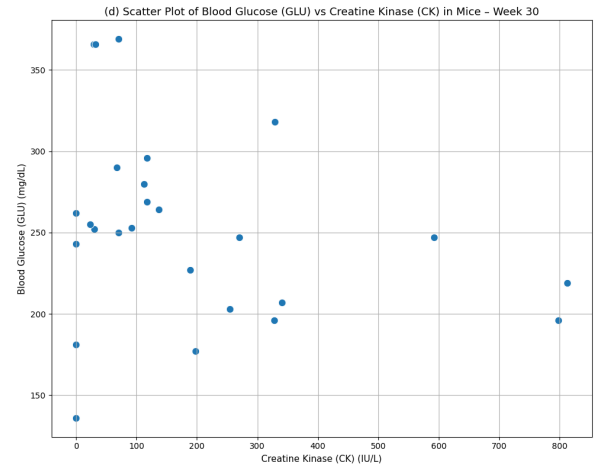


Figure 4.4: GLU vs CK (cov = -3739.74)

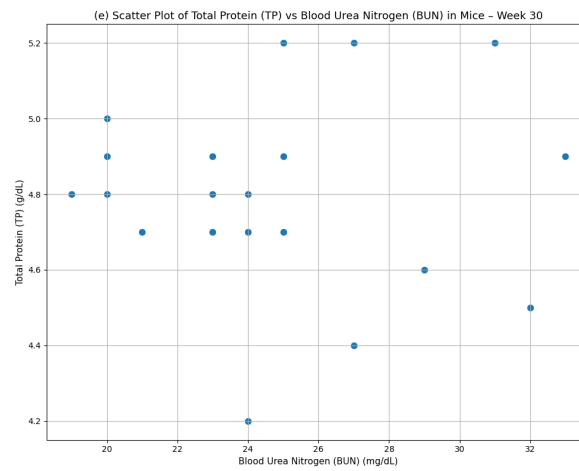


Figure 4.5: TP vs BUN (cov = 0.098)

NOTE: "cov" = covariance between the 2 plotted variables/observables.

A common observation across all the plots is the absence of any clear linear correlation between the variables, as evidenced by the scattered distribution of points in the scatter plots. Furthermore, attempting to fit the data using polynomial regression also does not appear to be feasible, as the data do not exhibit any discernible nonlinear trend.

However, this data clearly shows clustering:

- The rectangular region enclosed by (60,200), (100,200), (100,350) and (60,350) in Figure 4.1
- The rectangular region enclosed by (15,150), (30,150), (30,350) and (15,350) in Figure 4.2
- The rectangular region enclosed by (50,2.4), (100,2.4), (100,2.7) and (50,2.7) in Figure 4.3
- The rectangular region enclosed by (0,150), (400,150), (400,300) and (0,300) in Figure 4.4
- The rectangular region enclosed by (20,4.7), (26,4.7), (26,5.0) and (20,5.0) in Figure 4.5

In this case, covariance alone does not provide particularly meaningful insight, as its magnitude depends on the units and scale of the variables involved. A correlation-based measure, such as Spearman's correlation, would be more informative, as it evaluates the strength of a relationship independent of scale.

At present all that can be gauged is the fact that the covariance in case (e) is too small, indicating that the identified cluster might just be a visual artifact. Additionally in case (d) the covariance is negative indicating that GLU and CK are negatively correlated with each other, i.e: increase in one possibly leads to a decrease of the other.

Question 6

The histograms of the fat tissue mass distributions at week 8 and week 30 are shown below:

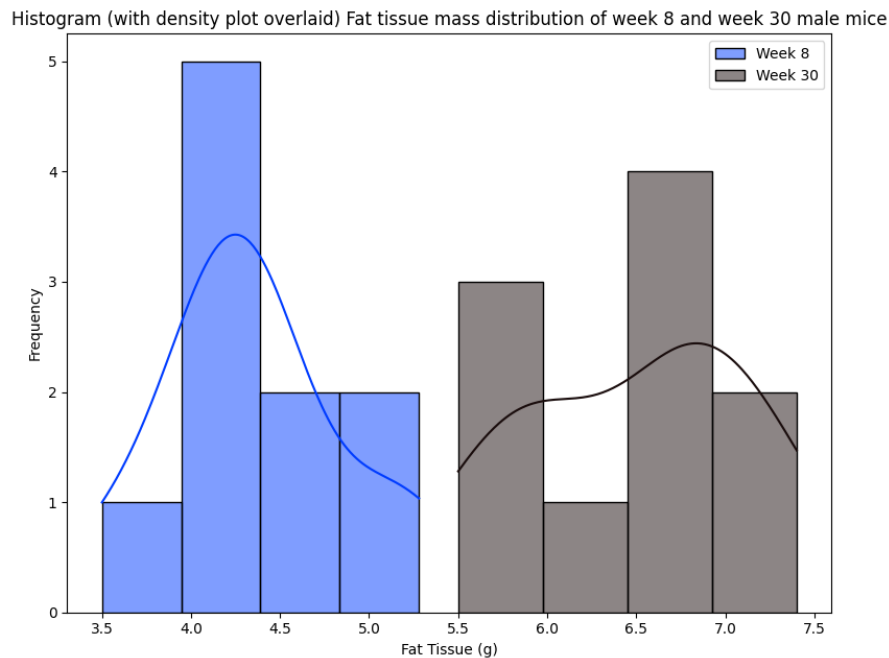


Figure 6: Fat tissue mass distribution at week 8 and week 30

Though these distributions do not look like normal distributions per se, but to evaluate whether sample means are different we are required to carry out a parametric test.

NOTE: Non-parametric Mann-Whitney U-test was performed and the output is given below.

Mann-Whitney U-Test Results:

Statistic: 0.0

P-value: 0.0001806347208075351

There is a significant difference in fat tissue mass between the male mice at week 8 and week 30.

F-Test (Comparing Significance of Variance)

An F-Test was performed to assess whether the variances differ significantly, in order to determine whether a separate-variance t-Test or a pooled-variance t-test should be used to compare the means.

H_0 : There is no significant difference in variance of fat tissue mass between the mice at week 8 and week 30 ($\sigma_{week_{30}} = \sigma_{week_8}$)

H_1 : There is a significant difference in variance of fat tissue mass between the mice at week 8 and week 30 ($\sigma_{week_{30}} \neq \sigma_{week_8}$)

The output of the program for F-Test is given below:

Variance of Week 30 = 0.36

Variance of Week 8 = 0.237

Calculated F statistic = 1.517

Critical F value = 3.179

Fail to reject the null hypothesis (p = 0.272)

Thus, There is no significant difference in variance of fat tissue mass between the mice at week 8 and week 30 in this study. Therefore a pooled variance t-test is carried out to compare the means.

Student t-test (Pooled Variance)

We carry out a one-tailed t-Test to assess whether mean of fat tissue mass significantly increased from week 8 to week 30.

H_0 : Fat tissue mass has not significantly increased from week 8 to week 30 in male mice ($\mu_{week_{30}} \leq \mu_{week_8}$)

H_1 : Fat tissue mass has significantly increased from week 8 to week 30 in male mice ($\mu_{week_{30}} > \mu_{week_8}$)

The output of the program for t-Test is given below:

```
Week 8 mean = 6.5
Week 30 mean = 4.354
Pooled Variance = 0.547

Calculated t value = 8.782
Critical t value  = 1.734

Reject the null hypothesis (p = 3.1659582844234535e-08)
```

Thus, fat tissue mass has significantly increased from week 8 to week 30 in male mice in this study. This is also clearly visible from the histogram plotted at the start.

Question 7

The histograms of the WDC count distribution at week 30 for male and female mice is shown below:

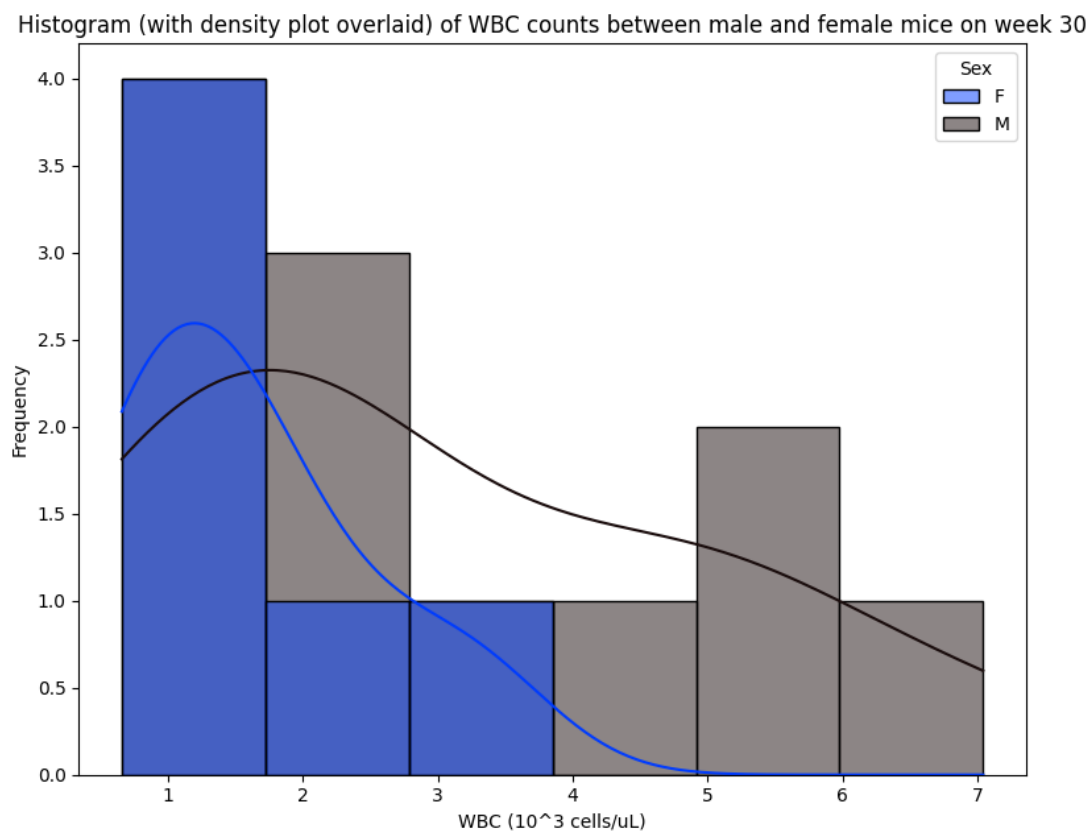


Figure 7: WDC count distribution at week 30 (Male and Female mice)

The given distributions are heavily right skewed and are not symmetric, as such we can infer the distributions are clearly non-normal, preventing us from using parametric testing.

Therefore we apply non-parametric testing (Mann-Whitney U-test) to compare the 2 samples.

Mann-Whitney U-Test

H_0 : Male and female mice at week 30 have the same WBC count.

H_1 : Male and female mice at week 30 do not have the same WBC count.

The output of the program for U-Test is given below:

Mann-Whitney U-Test Results:

Statistic: 50.5

P-value: 0.18955480737931263

There is no significant difference in WBC count between male and female mice on week 30.

There is not enough evidence to reject the null hypothesis as such we conclude that there is no statistically significant difference in WBC count between male and female mice on week 30.

Link to the google drive containing the python code used in this assignment:

https://drive.google.com/drive/folders/1UUF0k_-50DXR9J5LU0ddLIBhBFg-notw?usp=sharing